

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 8: Longest Common Subsequence

1. Aim

To find the Longest Common Subsequence (LCS) of two sequences using dynamic programming.

2. Objective

- Understand overlapping subproblems and optimal substructure.
- Construct an LCS using a dynamic programming table.
- Analyze time and space requirements.

3. Theory

The Longest Common Subsequence problem finds the longest sequence appearing in the same relative order in two sequences. For strings of lengths m and n , a dynamic programming table is constructed. If the current characters match, the diagonal value is increased by one; otherwise, the maximum of the top and left values is used. The standard solution has time and space complexity $O(mn)$.

4. Conclusion

Dynamic programming successfully finds the LCS by storing solutions of smaller prefixes and reusing them. The $O(mn)$ solution is useful in text comparison and sequence analysis.